**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 111 · Lesson 111 · OS Types, Boot Firmware, Drivers and VMs

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. In round robin, what happens when a running process's quantum expires but it is unfinished?

Q2. After pushing A then B onto a stack, which is popped first?

Q3. Why must a binary search operate on ordered data?

Q4. An OR gate has inputs 0 and 1. What is the output?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

real-time OS — System where meeting time constraints is essential.

BIOS/UEFI — Platform firmware that begins boot preparation.

hypervisor — Manages virtual machines running guest OSs.

device driver — Hardware-specific interface supplied to an OS.

## New learning

Match OS to use — Embedded/real-time emphasise constraints; multi-user and distributed coordinate shared work.

Boot and control — BIOS/UEFI firmware begins startup; drivers provide hardware-specific access.

Virtualise — A hypervisor runs guest OSs; a language VM executes intermediate code.

## Worked example

An industrial safety controller must respond within a guaranteed deadline. Choose an OS type and justify it.

1. Model answer — A hard real-time operating system is suited because missing a deadline is unacceptable; deterministic scheduling/latency matters more than high average throughput. An embedded implementation may also apply. — Check: link each claim to the scenario and use precise OCR terminology.

# We Do · Practise with guidance

A school wants to test Linux on Windows hardware and use a new printer. Explain the system software involved.

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Use a VM with a hypervisor to run Linux as a guest OS, while a printer driver exposes the printer to the host OS or guest when assigned. Boot firmware starts hardware initialisation before the main OS loads.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain two differences between multi-user and multitasking operating systems, giving examples. [6]

### Question 2

Describe the roles of BIOS/UEFI firmware and a device driver during startup and use. [6]

### Question 3

Compare a hypervisor virtual machine with a bytecode virtual machine and explain a real-time deadline. [8]

# Exit Ticket · Answer before checking

Every virtual machine must run a complete guest operating system.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. It returns to the ready queue (subject to scheduler rules).

Q2. B, because a stack is LIFO.

Q3. It discards half the search space using an ordering comparison.

Q4. 1.

### You Do pages · Independent questions

Q1. Multi-user supports separate simultaneous user sessions/accounts [3]; multitasking enables several processes to share CPU execution [3]; a system may do both.

Q2. Firmware initialises/checks hardware and begins booting [3]; device driver supplies OS-to-peripheral interface [3].

Q3. Hypervisor hosts virtualised OS/hardware [2]; bytecode VM executes intermediate instructions [2]; deadline determines correctness for hard real-time tasks [2]; appropriate context/trade-off [2].

### Exit Ticket · Model answer

False. Language virtual machines also execute intermediate code.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.